

Outlier Detection in Espresso Brewing

Camarena Mario *

Sauceda Oziel †

Ocaña Samantha ‡

Abstract

Espresso brewing is a precise process influenced by variables such as temperature, pressure, flow rate, and bean freshness. As these parameters fluctuate, they directly impact cup quality, creating a need for quantitative methods to ensure consistency and detect defective brews. Accordingly, this paper presents a data-mining model to identify anomalous espresso shots. Thirteen brewing logs recorded from a DE1 espresso machine were analyzed, each capturing the time-dependent behavior of pressure, flow, temperature, and water dispensed. The data were temporally aligned, standardized, and summarized using statistical features such as mean, maximum, and cumulative values. An L2-norm distance metric was then applied to compute an outlier score for each shot. The analysis successfully identified one shot that deviated across multiple parameters, particularly in pressure and flow profiles. The findings show that simple distance-based analysis, when combined with normalization and feature extraction, can reliably detect brewing inconsistencies and support data-driven quality control in espresso preparation.

1 Introduction

Espresso brewing is a precise process influenced by temperature, pressure, flow rate, and bean freshness. Small variations in these parameters can change how water moves through the compacted coffee bed, affecting extraction yield and the concentration of dissolved compounds in the final cup. The resulting differences are part of the extraction kinetics that define espresso quality. Maintaining a consistent cup requires adjusting grind size, dose, and temperature daily, a practice known as dialing in. Yet, even under controlled conditions, environmental factors such as humidity, bean aging, and machine variation continue to introduce variability.

Apiletti and Pastor [1] showed that espresso quality depends not only on mechanical parameters such as pressure and temperature but also on environmental conditions. They observed that humidity can affect grind behavior, requiring coarser settings in humid environments and finer ones in dry air to maintain the same extraction flow. This finding emphasizes that consistency in espresso preparation involves adapting to both machine conditions and surrounding climate. Similarly, Schmieder et al. [3] demonstrated that changes in flow rate, grind

level, and temperature significantly influence extraction kinetics, with finer grinds producing higher resistance and pressure. Their study using a DE1 espresso machine revealed how physical variables interact during brewing, underscoring the complexity of flow and pressure control.

Beyond machine and environmental effects, the condition of the coffee beans themselves plays a crucial role. Gantner et al. [2] found that storage temperature alters the volatile compound profile of roasted coffee, with lower storage temperatures preserving desirable sweet and chocolate aromas and higher temperatures increasing earthy and smoky notes. Although this work focused on sensory chemistry rather than extraction flow, it illustrates how bean aging and storage conditions can change the physical and chemical characteristics of coffee grounds. Such variations highlight how bean conditions can indirectly impact the brewing process and resulting flow dynamics.

Together, these studies [1, 2, 3] highlight the delicate balance of factors that determine espresso extraction. Even small deviations in grind, temperature, or bean freshness can create measurable differences in pressure and flow behavior. Accordingly, this study applies a data-driven approach to identify anomalous espresso shots using multivariate sensor data collected from a DE1 espresso machine. By resampling, normalizing, and analyzing time-series profiles of pressure, flow, temperature, and water dispensed, this work seeks to distinguish normal variability from meaningful deviations. The findings aim to clarify how differences in bean condition or process stability appear in sensor data, supporting a more interpretable, quantitative understanding of espresso brewing consistency.

2 Materials

The data that was being utilized was an espresso folder that contained 13 individual text files containing information about separate espresso shots. Categories from the data such as pressure, flow, basket/mix temperatures, resistance, and cumulative water dispensed were used primarily to analyze the sensor profiles. Other factors were not used as they did not represent areas of measured behavior and could affect the results of the outlier detection.

3 Methods

A major of the focus was on the brew phase. The order of steps is expressed as first keeping only the brew phase. With that it trimmed any of the warm up parts so that any non-brew data wouldn't mess or blur the real differences. Then it resampled each shot to have the same length. Espresso elapsed was utilized for this as the x-axis

*Department of Computer Science, University of Texas Rio Grande Valley, mario.camarena01@utrgv.edu

†Department of Computer Science, University of Texas Rio Grande Valley, oziel.sauceda01@utrgv.edu

‡Department of Computer Science, University of Texas Rio Grande Valley, samantha.ocana01@utrgv.edu

and to linearly interpolate each metric. Since shots contain various durations or sampling, this would construct curves that line up point by point. A standardized metric was used across all the shots otherwise known as Z-score, and for every metric it would subtract the total mean and then divide by the total standard deviation that was computed from all the aligned shots. This allowed the various metrics to contribute fairly.

In every metric it computed a mean for typical level, max for peak intensity, min for lowest level, and standard deviation to see how noisy it is. The sum presented the area under the curve across the aligned line. Flow and water dispensed showed the amount of liquid moving and pressure, temperature, and resistance captured the total magnitude across the shot. The rise time ranged from 10% - 90% for pressure and that represented how quick it rose. These computed the when and how much was comparable from all the shots across.

One number was assigned per shot for a total of 13 shots of espresso, and used to see how different a shot was from the others. The process of this was to make a table and for every shot, keep a small summary that contained information on the signals such as pressure, flow, basket/mix temp, water-dispensed, and resistance. These became the mean, max, and sum that represented one row for every shot. In addition to that features were placed on the same scale using a Z-score. This was done by centering the median and scale by the median absolute deviation for every column. This would construct a unit that could be compared across the various factors such as pressure, flow, and temperature that wouldn't be tricked by one value that was too extreme. Next, was combining all features into one distance, and that meant each shot would take the Euclidean length of the standardized value. The manner in seeing an outlier would be determined by the greater the distance, the farther it would be from the group and that would mean the shot value would be the most unusual. This is the determination of finding the outlier amongst the 13 shots.

4 Results

Figure 1. Summarizes all 13 espresso shot profiles across pressure, flow, basket temperature, mix temperature, water dispensed, and resistance. Most of the shots follow a similar repeatable pattern: pressure quickly ramps up and plateaus, flow decreases smoothly 10 seconds in as resistance builds, temperatures converge to a narrow band, water is dispensed almost linearly, and resistance peaks moderately before decaying. There is a clear, distinct visual difference between this consistent group and Shot 13.

Figure 2. Shows the outlier scores for all 13 shots, computed as the L2 norm of their standardized (z-scored) 39-dimensional feature vectors. Shot 13 has the highest score of 9.5227, which is 21.4% higher than the second highest

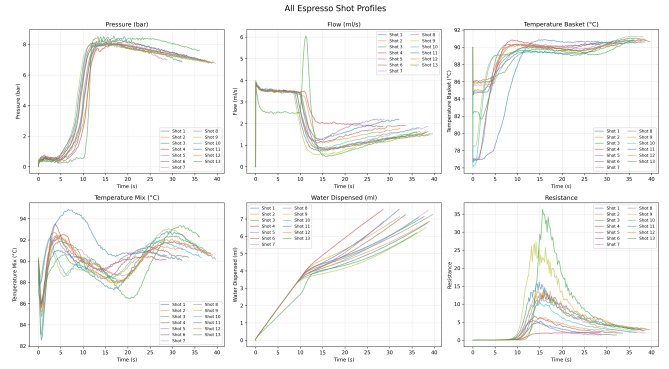


Figure 1

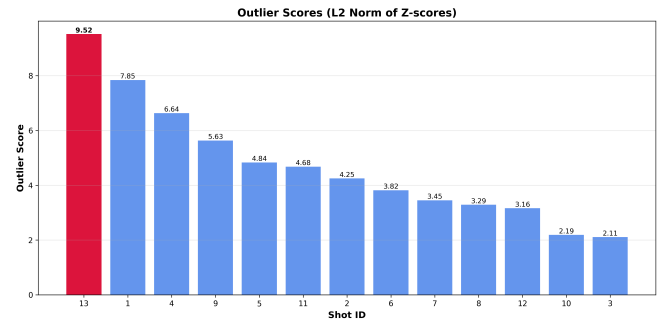


Figure 2

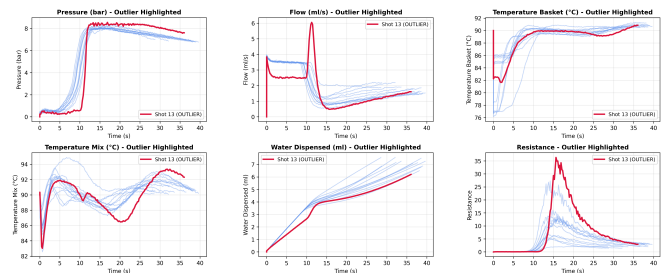


Figure 3

(Shot 1: 7.8459), 124.0% above the median (4.2518), and 2.19 standard deviations above the mean (4.7257), while all remaining shots cluster in a substantially lower range. This separation indicates that Shot 13 is uniquely distant from the typical brewing behavior.

Figure 3. Highlights Shot 13 (red) versus the remaining shots (blue) for each brewing variable, confirming that its high outlier score reflects consistent physical deviations rather than numerical noise. We observe a pronounced mid-shot flow spike, altered basket and mix temperature trajectories, reduced water dispensed, and the highest, most prolonged resistance peak, all supporting the conclusion that Shot 13 is the outlier.

5 Conclusion

This study developed and tested a simple anomaly detection pipeline for espresso brewing using sensor data collected from a DE1 espresso machine. The pipeline focused on the extraction phase of thirteen espresso shots and used descriptive features such as mean, maximum, and cumulative values of flow, pressure, temperature, and water volume to compare shots in a standardized way. By computing the L2 distance across all standardized variables, the method identified Shot 13 as the most distinct among the set. The analysis revealed that Shot 13 had a unique extraction pattern characterized by a sudden spike in flow rate, a delayed rise in pressure, and irregular temperature behavior. These combined deviations suggest differences in puck resistance or flow uniformity, which can occur when beans are older or when grind and tamping conditions change. The findings show that this approach can effectively isolate shots that behave differently from the rest of the dataset without needing labeled training data. The method is unsupervised and lightweight, relying only on the sensor data already collected by the DE1 machine. It does not require large datasets or re-training, making it suitable for practical use in café environments or automated quality-control systems. However, this approach was tailored to a small dataset and assumes a single anomaly. Future work could expand to larger datasets, integrate probabilistic or sequence-based models, and further examine how environmental factors such as humidity or bean freshness affect extraction behavior. Overall, the results demonstrate that feature-based L2 scoring on standardized brew variables can reliably detect abnormal shots. In this experiment, the pipeline successfully identified Shot 13 as the outlier consistent with expectations for a cup brewed using aged beans.

Acknowledgments

Special thanks to Dr. Wylie for hosting the event and Dr. Gao and Dr. Zhang for providing the problem! We are very tired, if this is incorrect our next guest is that the outlier is shot 3, if we were correct kindly ignore this and pretend this wasn't here. Thank you!!

References

- [1] D. Apiletti and E. Pastor. Correlating espresso quality with coffee-machine parameters by means of association rule mining. *Electronics*, 9(1):100, 2020. doi:10.3390/electronics9010100.
- [2] M. Gantner, et al. Effect of temperature and storage on coffee's volatile compound profile and sensory characteristics. *Foods*, 13(24):3995, 2024. doi:10.3390/foods13243995.
- [3] B. K. L. Schmieder, et al. Influence of flow rate, particle size, and temperature on espresso extraction kinetics. *Foods*, 12(15):2871, 2023. doi:10.3390/foods12152871.